John Carty

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Education

Swarthmore College, Swarthmore, PA — Expected May 2028

- B.S. Engineering; B.A. Astrophysics | GPA: 3.64/4.0
- Relevant Coursework: Analytical Dynamics, Thermofluid Mechanics, Linear Physical Systems Analysis, Stochastic & Numerical Methods, Electricity & Magnetism, Thermodynamics, Differential Equations

Work Experience

Plasma Physics & Systems Engineering Researcher | Swarthmore College — May 2026 – Present

- Designed, fabricated, and integrated a 1 kW LaB6 cathode assembly for the LiPD plasma device, increasing plasma density by approximately 100× over the previous cathode design
- Performed thermal FEA and electrical sizing of the 50 A heater, iterating emitter geometry and refractory supports to predict LaB6 temperatures above 2000 K within material limits
- Commissioned the completed assembly in vacuum and demonstrated sustained operation at the full 50 A design current while maintaining a stable argon discharge
- Developed and validated a Python full-orbit model using actual WHAM magnetic-field data and Boris integration, reproducing analytical magnetic-mirror confinement behavior
- Parallelized production sweeps on Swarthmore's Firebird HPC cluster, executing 100,000+ particle integrations and showing inferred radial loss fell from 19% to below 1% with a limiter-related flux-surface boundary

Biomimetic Robotics Researcher | Swarthmore College — Oct 2024 – Feb 2026

- Designed and fabricated a waterproof biomimetic swimming robot reproducing the coordinated motion of three pairs of barnacle nauplius limbs for low-Reynolds-number propulsion research with University of Washington collaborators
- Developed a compact clockwork gear-and-linkage mechanism in Fusion 360, matching scaled biological motion within 3% while reducing overall robot height by 20%
- Designed and integrated an adjustable light-activated motor-control circuit in LTspice, enabling contactless activation while submerged in viscous fluids

Physics Teaching Assistant | Swarthmore College — Fall 2025, Fall 2026

- Supported instruction for 80+ students in foundational courses for physics majors covering classical mechanics, optics, and quantum mechanics
- Led 30+ problem-solving sessions and provided one-on-one tutoring to more than 20 students

Projects

Reaction Wheel Attitude Control System — Mar 2026 – Present

- Developed Python and MATLAB simulations of single-axis spacecraft attitude dynamics using Euler's equations, analysis of the inertia tensor, and conservation of angular momentum
- Designed and 3D-printed an initial reaction-wheel testbed integrating an IMU and control electronics, establishing a platform for closed-loop testing against numerical predictions

Skills

Hardware & Testing: Electromechanical integration, vacuum/high-temperature hardware, experimental testing, 3D printing

Programming & Computation: Python (NumPy, SciPy, Matplotlib), MATLAB/Simulink, Java, Git, HPC/parallel computing

CAD & Analysis: SolidWorks, Fusion 360, Onshape, thermal FEA, LTspice